Amazon Reviews Sentiment Analysis Dashboard

Introduction

In today’s digital era, customer reviews play a crucial role in shaping product success on e-commerce platforms. This project focuses on analyzing Amazon customer reviews to extract meaningful insights about user satisfaction, product quality, and overall sentiment.

The main objective of this project is to understand customer behavior by analyzing ratings and textual reviews using sentiment analysis techniques. By combining data processing with interactive dashboards, this project helps in identifying trends, patterns, and areas of improvement for businesses.

Dataset Description

The dataset used in this project is the Amazon Customer Reviews Dataset, which contains detailed information about customer feedback.

Key Features of Dataset:

- Review Text (Customer feedback)
- Rating (1 to 5 stars)
- Review Date
- Reviewer Information

Data Preprocessing Steps:

- Removed missing and null values
- Converted data types (e.g., date format correction)
- Cleaned review text for analysis
- Handled inconsistent values (e.g., "1 review", "23 reviews")
- Filled missing values where necessary (e.g., NA for empty fields)

Tools & Technologies Used

- **Power BI** → Dashboard creation & visualization

- **Python** → Data cleaning and sentiment analysis
- **Pandas** → Data manipulation
- **NLTK & TextBlob** → Sentiment analysis

Power BI Visualizations & Insights

1) Cards (Overview Section)

This section displays key metrics such as:

- Total Reviews (~21K+)
- Average Rating
- Average Sentiment Score

Insight:
These KPIs provide a quick summary of overall dataset performance. Despite having a large number of reviews, the sentiment is mixed, indicating inconsistent customer experience.

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2) Sentiment Distribution (Donut Chart)

This visualization shows the proportion of:

- Positive Reviews (~48%)
- Negative Reviews (~44%)
- Neutral Reviews

Insight:
Although positive reviews are slightly higher, the negative reviews are also significantly high. This suggests that there is a strong need for product or service improvement.

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3) Sentiment Over Time (Line Chart)

This chart tracks how sentiment changes over time based on review dates.

Insight:
The sentiment shows a slight decline over time, which may indicate decreasing customer satisfaction or quality issues in recent periods.

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4) Rating Distribution (Pie Chart)

Displays count of ratings from 1-star to 5-star.

Insight:
A large number of users have given **1-star ratings**, which is a strong indicator of dissatisfaction. This highlights major issues with the product or service.

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5) Top Reviewers (Bar Chart)

Shows users who have given the highest number of reviews.

Insight:
Some users are highly active and contribute multiple reviews. These users can be valuable for identifying consistent feedback patterns.

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6) Sentiment vs Rating (Table Visualization)

Displays relationship between ratings and sentiment categories.

Insight:

- High ratings (4–5 stars) → Mostly Positive
- Low ratings (1–2 stars) → Mostly Negative

This confirms the expected correlation between rating and sentiment accuracy.

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7) Slicers / Filters

Used filters include:

- Review Date
- Rating
- Sentiment

Insight:
Slicers allow dynamic analysis, enabling users to explore data based on specific time periods or rating categories.

Key Insights Summary

- Total ~21K+ reviews were analyzed
- Majority of reviews are positive (~48%)

- Negative reviews are also high (~44%), indicating improvement scope
- Most users gave 1-star ratings → strong dissatisfaction signal
- Sentiment slightly declined over time
- Higher ratings align strongly with positive sentiment

Conclusion

This project demonstrates the power of combining data analysis, sentiment analysis, and visualization to understand customer feedback effectively. The findings reveal that while many users are satisfied, a significant portion of negative reviews highlights critical issues that need attention.

By leveraging these insights, businesses can improve product quality, enhance customer experience, and make data-driven decisions. Overall, this dashboard provides a comprehensive view of customer sentiment and serves as a valuable tool for business analysis.